Simulation of two-level quantum system using classical coupled oscillators

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Abstract

We show that the classical equations of motion of a coupled oscillator is equivalent to the Schrödinger's equation of a two-level quantum system. A simple experimental demonstration is presented to establish the analogy.

I. INTRODUCTION

Due to the difference in the underlying principles, quantum phenomenon is typically not expected to possess classical counterparts. However, though not all quantum phenomena have classical analogues, they are not uncommon. Classical analogues of quantum phenomenon serve as a source of additional insight, especially to those newly exposed to the subject. Nevertheless, the importance of these analogues is not merely pedagogical. Such analogues can be used for the formulation of new physical ideas[1]. Hence, they may even lead to further advancement of the theory. Even if such speculations are kept aside, such analogies have proved to be important, in areas like the design of mesoscopic devices at a macroscopic scale and quantum information processing[2]. There is no doubt that analogies provide a deeper grasp of the similarity between the classical and quantum world. Perhaps, they will also aid in better understanding the difference between the two. A detailed review of quantum-classical analogies is presented by Dragoman[3].

Two-level quantum systems are arguably the simplest non-trivial quantum systems and they are said to deviate most dramatically from classical systems[4]. However, there have been instances of demonstration of classical analogues of two-level quantum systems[5]. In this article, the mathematical analogy between the Schrödinger equation of a two level quantum system and the classical equations governing the motion of a coupled oscillator is explored. The results are compared with the dynamics of a two level quantum system, specifically a two level atom interacting with a monochromatic light field. Classical analogy between dressed states of coherent atom-laser interaction and the normal mode frequencies of a classical coupled oscillator has been explored previously[6]. Related phenomena, like electromagnetically induced transparency (EIT)[6,7] and double EIT[8] have been demonstrated in coupled electrical resonator circuits. We show here that, despite the simplicity of our candidate classical system, the similarity is mathematically sound. The analogy is also experimentally demonstrated, using a simple experimental setup suitable for undergraduates, and the results are compared to their theoretical quantum counterparts.

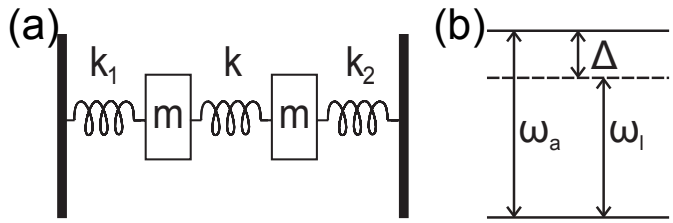

FIG. 1. (a) A classical coupled oscillator (b) Representation of a two level atom with atomic transition at ω_a interacting with monochromatic light field of frequency ω_l.

II. ANALOGY BETWEEN TWO-LEVEL QUANTUM SYSTEM AND CLASSICAL COUPLED OSCILLATOR

Consider a pair of coupled oscillators as shown in figure 1, consisting of two equal masses of magitude m each, attached to opposite walls with springs of spring constants k_1 and k_2, and attached to each other by a coupling spring of spring constant k. If the displacement of the masses from their equilibrium positions are denoted by x_1 and x_2, application of Newton's laws yields their equations of motion as

$$-\frac{d^2}{dt^2}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \mathcal{V}_1^2 & -\Omega^2 \\ -\Omega^2 & \mathcal{V}_2^2 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \tag{1}$$

where $\mathcal{V}_1 = \sqrt{\frac{k_1+k}{m}}$ and $\mathcal{V}_2 = \sqrt{\frac{k_2+k}{m}}$ are the natural frequencies of each of the oscillators when the other mass is held fixed and $\Omega = \sqrt{\frac{k}{m}}$ is indicative of the coupling strength between the oscillators. We denote the square matrix on the righthand side of equation (1) by $\mathbf{A}$. The square roots of the two eigenvalues of $\mathbf{A}$, denoted by ω_1 and ω_2, are

$$\begin{aligned} \omega_1 &= \left(\frac{\mathcal{V}_1^2+\mathcal{V}_2^2}{2} + \frac{1}{2}\sqrt{\left(\mathcal{V}_1^2-\mathcal{V}_2^2\right)^2+4\Omega^4}\right)^{\frac{1}{2}} \\ \omega_2 &= \left(\frac{\mathcal{V}_1^2+\mathcal{V}_2^2}{2} - \frac{1}{2}\sqrt{\left(\mathcal{V}_1^2-\mathcal{V}_2^2\right)^2+4\Omega^4}\right)^{\frac{1}{2}} \end{aligned} \tag{2}$$

The motion of the two masses, in general, will be a superposition of the two normal modes of vibration, whose frequencies are given by ω_1 and ω_2[9]. The general equations of motion are :

$$\begin{aligned} x_1(t) &= f_1 a_{11} e^{-i(\omega_1 t+\phi_1)} + f_2 a_{21} e^{-i(\omega_2 t+\phi_2)} \\ x_2(t) &= f_1 a_{12} e^{-i(\omega_1 t+\phi_1)} + f_2 a_{22} e^{-i(\omega_2 t+\phi_2)} \end{aligned} \tag{3}$$

where $\begin{bmatrix} a_{11} \\ a_{12} \end{bmatrix}$ and $\begin{bmatrix} a_{21} \\ a_{22} \end{bmatrix}$ are the normalised eigen vectors of $\mathbf{A}$ corresponding to ω_1 and ω_2 respectively. Given the initial positions and velocities of the two oscillators, f_1, f_2, ϕ_1 and ϕ_2 can be determined uniquely.

Equation (1) may be rewritten as

$$\left(i\frac{d}{dt}\right)^2 \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \left(\sqrt{\mathbf{A}}\right)^2 \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$\implies i\hbar\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \hbar\sqrt{\mathbf{A}} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \tag{4}$$

It may be noted that equation (4) is exactly of the form of Schrödinger's equation governing a two level quantum system where $\hbar\sqrt{\mathbf{A}}$ serves as the Hamiltonian of the system. Since $\mathbf{A}$ is a symmetric and positive definite matrix, it can be diagonalised by a similarity transformation, $\mathbf{P^{-1}AP}$. The roots of the diagonal elements can be taken and the resulting matrix can be transformed back to obtain $\sqrt{\mathbf{A}}$. The transformation matrix $\mathbf{P}$ is given by $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ where $\theta = \frac{1}{2}\tan^{-1}\left(\frac{2\Omega^2}{\mathcal{V}_1^2-\mathcal{V}_2^2}\right)$. By following the above method, $\sqrt{\mathbf{A}}$ can be determined and equation (4) can be written as

$$i\hbar\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \frac{\hbar}{2} \begin{bmatrix} \omega_0 - \Delta & \Omega_0 \\ \Omega_0 & \omega_0 + \Delta \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \tag{5}$$

where $\omega_0 = (\omega_1 + \omega_2)\cos 2\theta$, $\Delta = (\omega_1 - \omega_2)\cos 2\theta$ and $\Omega_0 = (\omega_1 - \omega_2)\sin 2\theta$.

Equation (5) can be compared to the quantum mechanical equations of motion of a two level quantum system, specifically to that in which a monochromatic laser beam is interacting with a two level atom close to resonance[10], as shown in figure 1. Δ would then denote the detuning, which is the difference between the laser frequency and the resonance frequency of the atomic transition line. Ω_0 would be the Rabi oscillation frequency, the frequency at which the population of the atomic states oscillate for laser frequency exactly at resonance. For non-zero detuning, the population oscillates with effective Rabi frequency, $\Omega_R = \sqrt{\Omega_0^2 + \Delta^2}$. Also, the transition probability to the excited state is Ω_0^2/Ω_R^2.

In the case of the coupled oscillators, it can be shown using equation (2) that $\omega_1 + \omega_2 \approx \mathcal{V}_1 + \mathcal{V}_2$ for $\Omega \ll \omega_1, \omega_2$. Using this approximation, it can be shown that detuning, $\Delta \approx \mathcal{V}_1 - \mathcal{V}_2$

and Rabi frequency, $\Omega_0 = \frac{2\Omega^2}{\nu_1+\nu_2}$. The effective Rabi frequency for finite detuning then becomes $\Omega_R = \omega_1 - \omega_2 = \sqrt{\Delta^2 + \Omega_0^2}$. Also, note that the square of the amplitudes of the oscillators normalised to the total energy of the system can be compared to the probability amplitudes of the population of the corresponding atomic states.

In the case of laser atom interaction, Δ can be varied by changing the laser frequency and the Rabi frequency Ω_0 remains constant for a fixed amplitude of laser field. However, in the case of the coupled oscillators, Δ can only be varied by changing the spring constant of one of the oscillators. As a result, the Rabi frequency also changes, since it depends on the natural frequencies of the oscillators. However, the Δ dependence of the Rabi frequency can be suitably implemented in the analysis for the correct comparison of both the systems.

III. EXPERIMENTAL DEMONSTRATION OF THE ANALOGY

A schematic diagram of the experimental setup used to demonstrate the above analogy is shown in figure 2. Two rulers R_1 and R_2 are mounted vertically, side by side, on an optical table. When struck at the top end, the rulers vibrate with their natural frequencies, with the tensile property of their material accounting for the spring constant. A rubber band is put around them to couple the oscillators. The spring constant of the rubber band along with the distance of the rubber band from the base of the rulers decide the strength of the coupling and hence the Rabi frequency. In our experiment, we chose a suitable Rabi frequency by changing the position of the rubber band.

The arrangement for quantitative measurement of the oscillations is also shown in figure 2. A laser beam transmitted through a polarising cube beam splitter (PBS) and a quarter wave plate is retro-reflected by a mirror mounted on the ruler R_2. The retro-reflected beam is reflected by the same cube beam splitter to fall on the quad-detector D_2. The reflected beam from the first cube beam splitter is again reflected by a second polarising cube beam splitter and transmitted through a quarter wave plate. This beam is similarly retro-reflected by a mirror mounted on the ruler R_1 and transmitted through the second cube beam splitter to fall on the quad detector D_1. The quad detectors have four photo-diodes arranged as four quarters of a circle. When the rulers are at rest, these two reflected beams are made to fall at the centre of the quad detectors. The vertical shift in the reflected beam due to the vibration of the ruler causes a difference in the intensity of light falling on either half of the

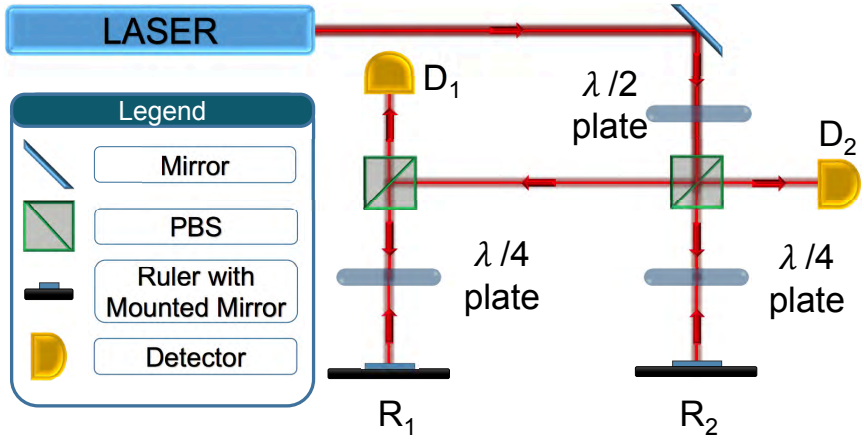

FIG. 2. Schematic diagram of the experimental setup.

photodiode. The movement of the rulers are traced by amplifying the potential difference between the two halves of the quad detectors and feeding it into an oscilloscope. A typical signal at near resonant condition, that is $\mathcal{V}_1 \approx \mathcal{V}_2$ or $\Delta \approx 0$, is shown in figure 3. As in the case of Rabi oscillation of the population of a two level atom interacting with a laser beam, the oscillation amplitudes of the two oscillators are seen to alternate, one showing maximum oscillation amplitude while the other is showing minimum. This 'transfer' of oscillation amplitude, which is equivalent to the transfer of energy, is observed to be maximum at $\Delta = 0$, where the transfer is nearly complete.

For the quantitative measurements of the analogues of different features of the Rabi oscillation in these coupled oscillators, the following procedure was adopted. By preventing the motion of one ruler and only striking the other the natural frequencies of the oscillators, $\mathcal{V}_1$ and $\mathcal{V}_2$, were determined by fitting the resulting signals with the equation of a damped harmonic oscillator. The natural frequency $\mathcal{V}_1$ of oscillator R_1 was changed by sticking small weights to it, while keeping $\mathcal{V}_2$ fixed. For each frequency $\mathcal{V}_1$, the coupled oscillations of the oscillators, like that in figure 3, were recorded and fitted with equations (3) to determine the normal mode frequencies ω_1 and ω_2. Treating $\mathcal{V}_1$ as the independent parameter and $\mathcal{V}_2$ and Ω as the fitting parameters, the normal mode frequencies ω_1 and ω_2 were fitted with the expressions in equation (2). The normal mode frequencies as a function of detuning Δ is shown in figure 4. The typical *avoided crossing* seen in atomic transition lines near resonance can be observed in the case of the classical coupled oscillators. Ω, which is related to the

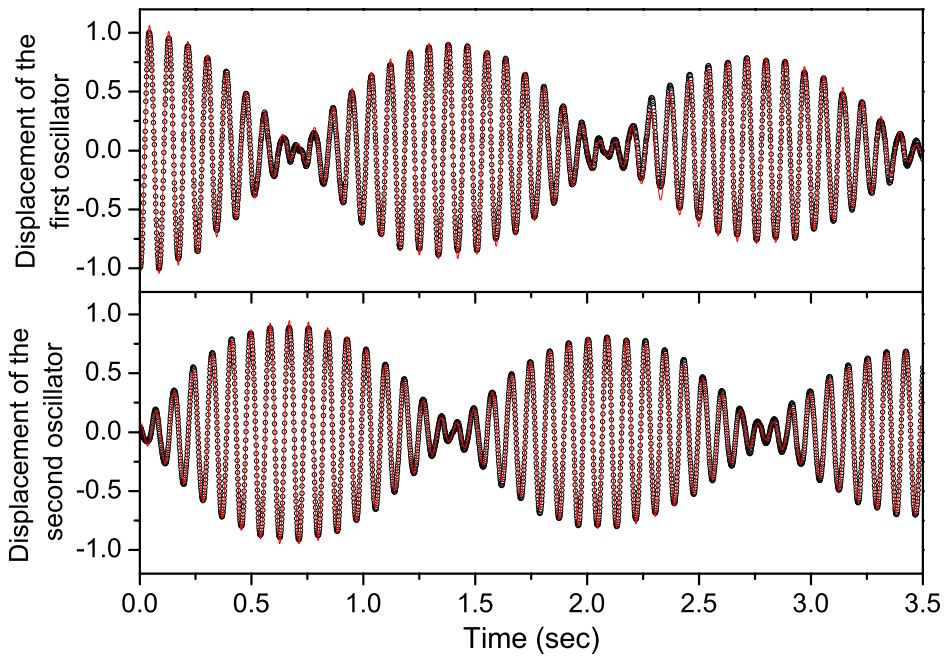

FIG. 3. The normalized displacement of the rulers from their mean positions as a function of time. Experimental data is shown as black open circles. The red solid lines are the curves fitted using the real part of the equation (3).

spring constant of the coupling spring, was determined from the fitting. Given a natural frequency and knowing Ω, the Rabi frequency Ω_0 can be determined using $\Omega_0 = \frac{2\Omega^2}{\mathcal{V}_1+\mathcal{V}_2}$. The effective Rabi frequency Ω_R, which is the rate at which energy is transferred between the oscillators, is determined using $\sqrt{\Omega_0^2 + \Delta^2}$ and plotted as a function of Δ in the inset of figure 4.

In the case of a two level atom interacting with a laser field, the transition probability of the atom to the excited state is Ω_0^2/Ω_R^2. This is equivalent to the percentage of the energy transferred from one oscillator to the other. For quantitative analysis, curves similar to those in figure 3 were fitted with the real part of the equation (3). The energy of R_1 is proportional to $|x_1|^2 = |f_1 a_{11}|^2 + |f_2 a_{21}|^2 + 2 f_1 f_2 a_{11} a_{21} \cos(\Omega_R t + (\phi_1 - \phi_2))$. $f_1 a_{11}$, $f_2 a_{21}$, ϕ_1 and ϕ_2 can be determined from the fit. Given the initial condition that the ruler R_1 is only struck, $|x_1|^2 = |f_1 a_{11}|^2 + |f_2 a_{21}|^2 + 2 f_1 f_2 a_{11} a_{21}$ at $t = 0$, which is maximum. At time, $t = \pi/\Omega_R$, the amplitude of R_1 becomes minimum and the amplitude of R_2 becomes maximum. The percentage of maximum energy transferred to R_2 is then proportional to $\left(|x_1|^2_{max} - |x_1|^2_{min}\right) / |x_1|^2_{max}$. This normalised energy transfer as function Δ is shown in

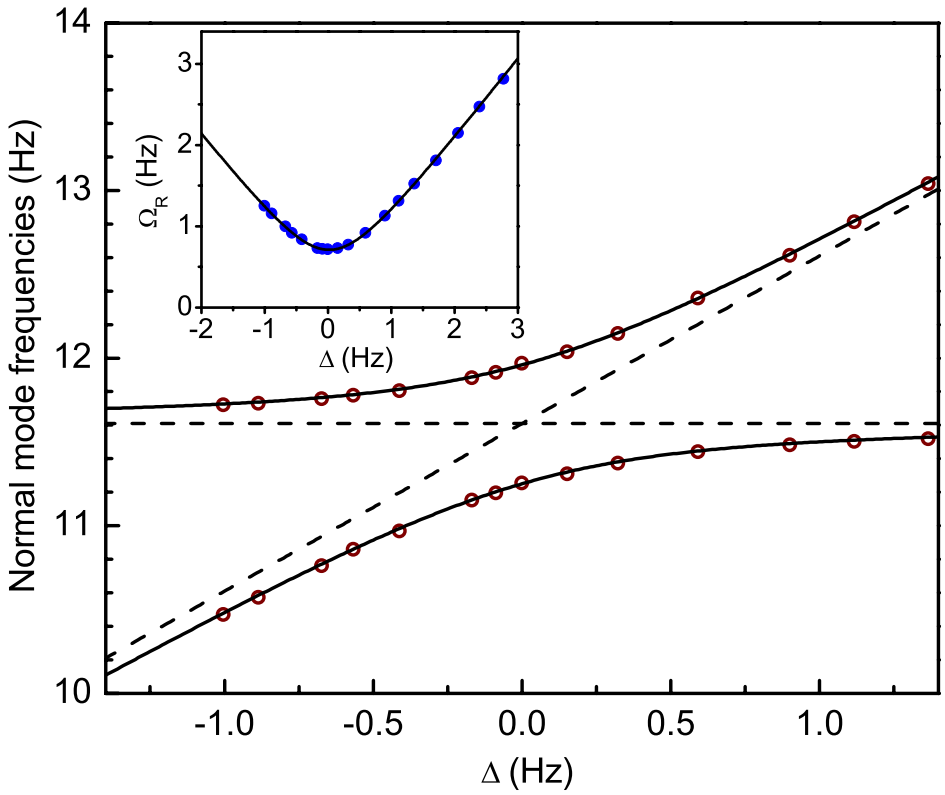

FIG. 4. Normal mode and natural frequencies as function of detuning. The open circles are the experimental data and the solid lines are the fitted curve as described in the text. The dotted lines are the natural frequencies of both the oscillators. (Inset) Effective Rabi frequency as a function of Δ. The closed circles are the difference of the normal mode frequencies and the solid line is for $\Omega_R = \sqrt{\Omega_0^2 + \Delta^2}$.

figure 5.

With the above initial conditions, the solutions of the coupled oscillator system indicates that the amplitude of oscillation of R_2 will go to zero at its minimum. However, the deviation from this behaviour was observed, probably due to non-perfect initial condition or the coupling of energy to other mechanical pieces in the setup. The ratio of the square of the minimum amplitude to the square of the maximum amplitude of R_2 is taken as relative error in our experimental data.

The experimental data compares with the theoretical plot of Ω_0^2/Ω_R^2 as shown by the dotted line in figure 5. A systematic shift of the experimental data from the theoretical model can be seen. We infer this systematic shift to be due to the additional masses added to R_1 to change its natural frequency. Due to the additional mass δm, the natural frequency

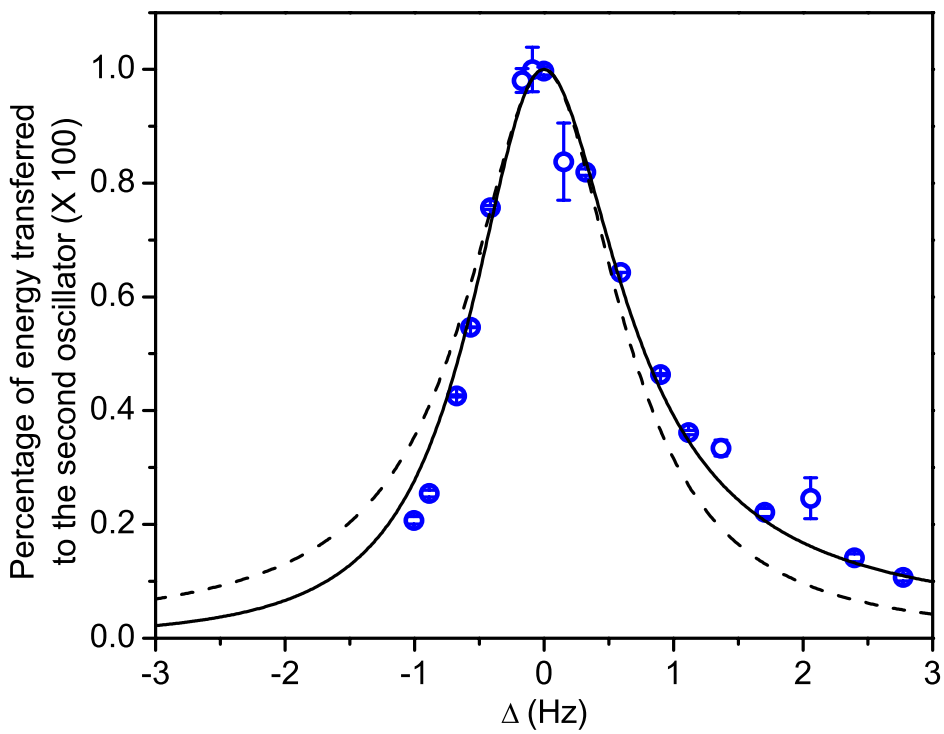

FIG. 5. Percentage of energy transferred from ruler R_1 to R_2. The open circles with the error bar are the experimental data. The dotted (solid) lines are for Ω_0^2/Ω_R^2 without (with) correction in the coupling strength due to additional mass in the ruler R_1 to change it's natural frequency.

$\mathcal{V}_1$ changes as $\sqrt{(k_1+k)/(m+\delta m)}$ and at the same time, the coupling strength Ω changes as $\sqrt{k/(m+\delta m)}$. Hence, the coupling strength should be corrected for each Δ instead of keeping it constant. However, the corrected coupling strength can approximately be calculated for small additional mass for each Δ as $\Omega_c = \Omega + \Delta\left(\mathcal{V}_1+\mathcal{V}_2\right)/\omega_2^2$. Then, the percentage of energy transfer from R_1 to R_2 calculated using the corrected Rabi frequency, $\Omega_0 = 2\Omega_c^2/(\mathcal{V}_1+\mathcal{V}_2)$, is seen to match better with the experimental data, as shown by the solid line in figure 5.

We would like to mention here that such an analysis of correction to the Rabi frequency due to additional mass on R_1 is not perfect. Due to the additional mass on R_1, the coupling strengths relevant in transferring energy from R_1 to R_2 and R_2 to R_1 becomes $\sqrt{k_1/(m+\delta m)}$ and $\sqrt{k_1/m}$ respectively. As a result, the matrix $\mathbf{A}$ in equation (1) will not be symmetric. Hence, $\sqrt{\mathbf{A}}$ cannot be compared to the Hamiltonian of a two level quantum system. However, if $\Delta, \Omega_0 \ll \mathcal{V}_1, \mathcal{V}_2$, the correction is small and the analogy holds better, which is evident in figure 5. Complete analogy can ideally be demonstrated by performing the experiment by

changing only k_1, keeping k and m fixed. In this case, the coupling between the oscillators is symmetric and hence $\sqrt{\mathbf{A}}$ will remain symmetric for any detuning. In our experiment, it was difficult to change k_1 as it depended on the tensile property of the ruler. It was observed that changing the coupling of R_1 by coupling it with a rigid wall resulted in larger damping, which induced larger errors in the experimental data.

IV. CONCLUSION

The analogy between classical coupled oscillators and two level quantum system has been established mathematically. Some aspects of the analogy were demonstrated using a simple experimental setup suitable for undergraduates. Such mathematical exercises and experimental demonstrations could aid students in the development of intuition in the non-intuitive subject of quantum mechanics. The analogies can also be further extended to study phenomena associated with multi-level quantum systems by increasing the number of oscillators in the coupled oscillator system. It may also lead to the formulation of new physical ideas. For example, quantum wave function does not have a direct physical interpretation and is connected to the physical world only through the probability interpretation. However, in case of this classical counterpart, the analogue of the quantum wave function is the oscillatory motion of the coupled oscillator which can directly be observed, as seen in figure 3. Hence, such an analogy may be useful in developing a physical intuition of the quantum wave function directly, while keeping the probability interpretation intact.

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